

ARTIFICIAL INTELLIGENCE

B.Sc. I.T(Sem- V)

Book Name:- **A Modern Approach**
by Stuart J. Russell and Peter Norvig

Publisher:- Pearson

----- Third Edition-----

UNIT-1

Chapter- 1 :- Introduction:-

- What is Artificial Intelligence?
- Foundations of AI,
- history,
- the State of AI today.

Chapter- 2:- Intelligent Agents :-

- agents and environment,
- good behavior,
- nature of environment,
- the structure of agents.

Introduction of AI

- Artificial= " man made"
- Intelligence= "thinking power"
- It is branch of Computer Science by which we can create intelligent machine which can behave like a human, think like humans and able to make decisions.

Defination of AI:-

- AI is concerned with the design of intelligence in artificial devices.
- So building intelligence into man-made systems is known as AI.
- Intelligence is a property in which machine behaves in best possible manner
- Intelligence is something which characterizes human.

Applications of AI:-

- Computer Vision
- AI in gaming
- Image Recognition
- Robotics (Sophia)
- NLP
- Speech Recognition
- AI in Agriculture

Weak AI and Strong AI:-

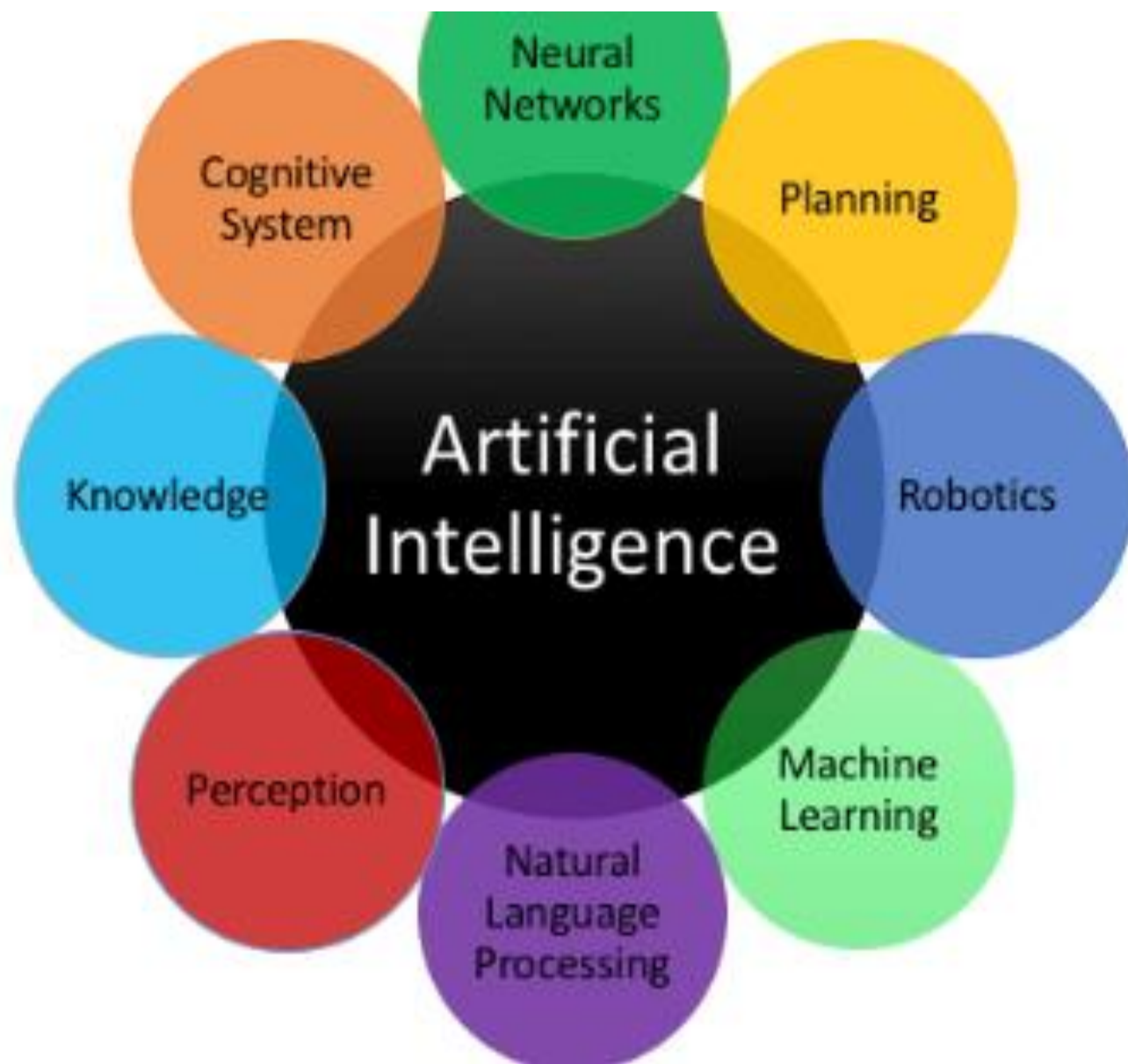
- Weak AI:- It deals with creation of computer based AI that cannot truly reason and solve problems but can act as if it were intelligent (Simulating Intelligent)
- Strong AI:- It deals with cognitive mental status that can think, reason and understand the behavior .

Introduction:-

What is Artificial Intelligence?

- According to the father of Artificial Intelligence, John McCarthy, it is *“The science and engineering of making intelligent machines, especially intelligent computer programs”*.
- Artificial Intelligence is a way of **making a computer, a computer-controlled robot, or a software think intelligently**, in the similar manner the intelligent humans think.
- AI is accomplished by studying how human brain thinks, and how humans learn, decide, and work while trying to solve a problem, and then using the outcomes of this study as a basis of developing intelligent software and systems.

The definitions of AI on the top are concerned with **thought processes and reasoning**, whereas the ones on the bottom address **behavior**.



Views of AI fall into four categories:

There are four different AI approaches based on focus and performance perspectives

PERFORMANCE MEASUREMENT	
vs. Humans	
vs. Ideal (Rationality)	
FOCUS	Thought Process/ Reasoning
	Behavior
METHODS	Based on human physiology
	Based on mathematics and logic

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Acting humanly: The Turing Test Approach

- **natural language processing** to enable it to communicate successfully in English (or some other human language);
- **knowledge representation** to store information provided before or during the interrogation;
- **automated reasoning** to use the stored information to answer questions and to draw new conclusions;
- **machine learning** to adapt to new circumstances and to detect and extrapolate patterns.

To pass the total Turing Test, the computer will need -----

- computer **vision** to perceive objects, and
 - **robotics** to move them about.
-
- Turing (1950) "Computing machinery and intelligence":
 - "Can machines think?" → "Can machines behave intelligently?"
 - Operational test for intelligent behavior: the Imitation Game
 - Predicted that by 2000, a machine might have a 30% chance of fooling a lay person for 5 minutes
 - Anticipated all major arguments against AI in following 50 years
 - Suggested major components of AI: knowledge, reasoning, language understanding, learning.

Thinking humanly: cognitive modeling

If we are going to say that a given program thinks like a human, we must have some way of determining how humans think. We need to get *inside* the actual workings of human minds. There are two ways to do this: through introspection--trying to catch our own thoughts as they go by--or through psychological experiments.

1960s "cognitive revolution": information-processing psychology

- Requires scientific theories of internal activities of the brain
- -- How to validate? Requires
 - 1) Predicting and testing behavior of human subjects (top-down)
 - or 2) Direct identification from neurological data (bottom-up)
- Both approaches (roughly, Cognitive Science and Cognitive Neuroscience) are now distinct from AI

Thinking rationally: "laws of thought"

The Greek philosopher Aristotle was one of the first to attempt to codify "right thinking," that is, irrefutable reasoning processes. His famous **sylogisms** provided patterns for argument structures that always gave correct conclusions given correct premises. For example, "Socrates is a man; all men are mortal; therefore Socrates is mortal." These laws of thought were supposed to govern the operation of the mind, and initiated the field of **logic**.

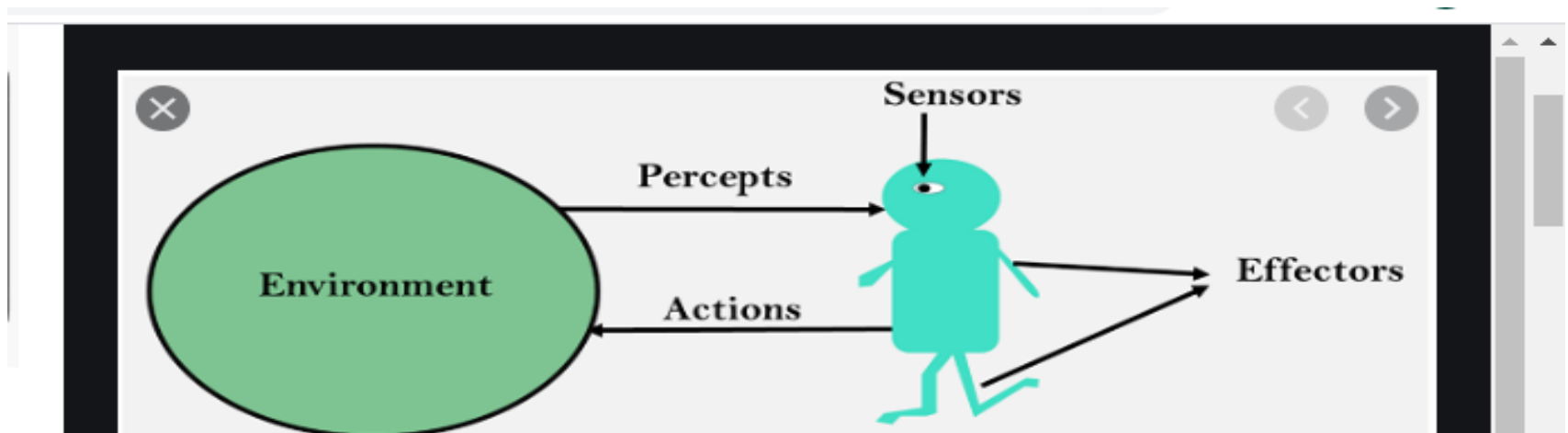
- Aristotle: what are correct arguments/thought processes?
- Several Greek schools developed various forms of *logic: notation* and *rules of derivation* for thoughts; may or may not have proceeded to the idea of mechanization
- Direct line through mathematics and philosophy to modern AI
- Problems:
 1. Not all intelligent behavior is mediated by logical deliberation
 2. What is the purpose of thinking? What thoughts should I have?

Acting rationally: rational agent

- **Acting rationally** means **acting** so as to achieve one's goals, given one's beliefs. An **agent** is just something that perceives and acts. In the logical approach to AI, the emphasis is on correct inferences.
- **Rational** behavior: doing the right thing
- The right thing: that which is expected to maximize goal achievement, given the available information
- Doesn't necessarily involve thinking – e.g., blinking reflex – but thinking should be in the service of rational action

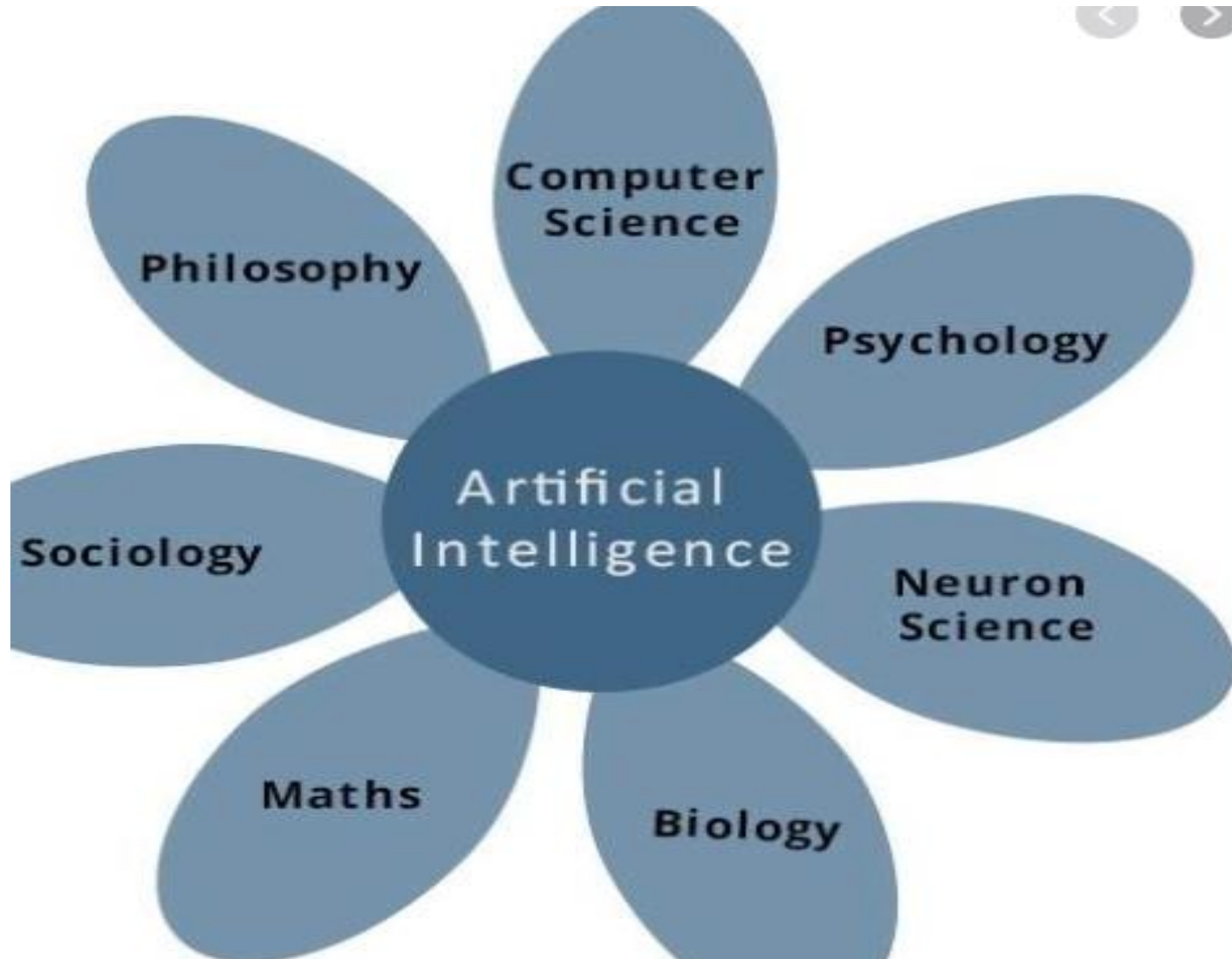
Rational agents

A **rational agent** is an **agent** which has clear preference, models uncertainty, and acts in a way to maximize its performance measure with all possible actions. A **rational agent** is said to perform the right things.



- An **agent** is an entity that perceives and acts
- This course is about designing rational agents
- For any given class of environments and tasks, we seek the agent(or class of agents) with the best performance

2) FOUNDATIONS OF ARTIFICIAL INTELLIGENCE



FOUNDATIONS OF ARTIFICIAL INTELLIGENCE

AI Is Not Alone at Home



□ Crossbreeding of a lot of fields

Philosophy	Logic, methods of reasoning, mind as physical system, foundations of learning, language, rationality.
Mathematics	Formal representation and proof, algorithms, computation, (un)decidability, (in)tractability
Statistics	Modeling uncertainty, learning from data
Economics	Utility, decision theory, rational economic agents
Neuroscience	Neurons as information processing units
Psychology / Neuro Science	How do people behave, perceive, process cognitive information, represent knowledge
Computer Engineering	Building fast computers
Control Theory	Design systems that maximize an objective function over time
Linguistics	Knowledge representation, grammars

FOUNDATIONS OF ARTIFICIAL INTELLIGENCE

1. **Philosophy:-**

- Can formal rules be used to draw valid conclusions?
 - How does the mind arise from a physical brain?
 - Where does knowledge come from?
 - How does knowledge lead to action?
-
- Aristotle (384–322 B.C.)-- was the first to formulate a precise set of laws governing the rational part of the mind.
 - Ramon Lull (d. 1315) had the idea that useful reasoning could actually be carried out by a mechanical artifact.
 - Thomas Hobbes (1588–1679) proposed that reasoning was like numerical computation, that “we add and subtract in our silent thoughts.”
 - Around 1500, Leonardo da Vinci (1452–1519) designed, but did not build a mechanical calculator.

- The first known calculating machine was constructed around 1623 by the German scientist Wilhelm Schickard (1592–1635), although the Pascaline, built in 1642 by Blaise Pascal (1623–1662).
- Gottfried Wilhelm Leibniz (1646–1716) built a mechanical device intended to carry out operations on concepts rather than numbers but its scope was rather limited.
- In 1651 book *Leviathan*, Thomas Hobbes suggested the idea of an “artificial animal,” arguing “For what is the heart but a spring; and the nerves, but so many strings; and the joints, but so many wheels.”
- Ren e Descartes (1596–1650) gave the first clear discussion of the distinction between mind and matter and of the problems that arise.
- Descartes was a strong advocate of the power of reasoning in understanding RATIONALISM the world, a philosophy now called **rationalism**.
- Descartes was also a proponent of **dualism**.

He held that there is a part of the human mind (or soul or spirit) that is outside of nature, exempt from physical laws. Animals, on the other hand, did not possess this dual quality; they could be treated as machines.

- An alternative to dualism is **materialism**, which holds that the brain’s operation according to the laws of physics *constitutes* the mind.
- The **empiricism** movement, starting with Francis Bacon’s (1561–1626) *Novu Organum*,² is characterized by a dictum of John Locke (1632–1704): “Nothing is in the understanding, which was not first in the senses.”

2. Mathematics

- What are the formal rules to draw valid conclusions?
- What can be computed?
- How do we reason with uncertain information?
- In 1879, Gottlob Frege (1848–1925) extended Boole's logic to include objects and relations.
- Alfred Tarski (1902–1983) introduced a theory of reference that shows how to relate the objects in a logic to objects in the real world.
- David Hume's (1711–1776) *A Treatise of Human Nature* (Hume, 1739) proposed what is now known as the principle of **induction**: that general rules are acquired by exposure to repeated associations between their elements
- Building on the work of Ludwig Wittgenstein (1889–1951) and Bertrand Russell (1872–1970), the famous Vienna Circle, led by Rudolf Carnap (1891–1970), developed the doctrine of **logical positivism**.
- This doctrine holds that all knowledge can be characterized by logical theories connected, ultimately, to **observation sentences** that correspond to sensory inputs; thus logical positivism combines rationalism & empiricism.
- The **confirmation theory** of Carnap and Carl Hempel (1905–1997) attempted to analyze the acquisition of knowledge from experience.
- Carnap's book *The Logical Structure of the World* (1928) defined an explicit computational procedure for extracting knowledge from elementary experiences.

- The first nontrivial **algorithm** is thought to be Euclid's algorithm for computing greatest common divisors.
- In 1930, Kurt Godel (1906–1978) showed that there exists an effective procedure to prove any true statement in the first-order logic of Frege and Russell.
- In 1931, Godel showed that limits on deduction do exist. His **incompleteness theorem** showed that in any formal theory as strong as Peano arithmetic (the elementary theory of natural numbers), there are true statements that are undecidable in the sense that they have no proof within the theory.
- Alan Turing (1912–1954) tried to characterize exactly which functions *are* **computable**—capable of being computed.
- Although decidability and computability are important to an understanding of computation, the notion of **tractability** has had an even greater impact.
- The theory of **NP-completeness**, pioneered by Steven Cook (1971) and Richard Karp (1972), provides a method. Cook and Karp showed the existence of large classes of canonical combinatorial search and reasoning problems that are NP-complete.

Besides logic and computation, the third great contribution of mathematics to AI is the theory of **probability**.

3) Economics:-

- How should we make decisions so as to maximize payoff?
- How should we do this when others may not go along?
- How should we do this when the payoff may be far in the future?
- The science of economics got its start in 1776, when Scottish philosopher Adam Smith (1723–1790) published *An Inquiry into the Nature and Causes of the Wealth of Nations*.
- The mathematical treatment of “preferred outcomes” or **utility** was first formalized by L’eon Walras (pronounced “Valrasse”) (1834-1910) & was improved by Frank Ramsey (1931) and later by John von Neumann and Oskar Morgenstern in their book *The Theory of Games and Economic Behavior* (1944).
- **Decision theory**, which combines probability theory with utility theory, provides a formal and complete framework for decisions.
- Von Neumann and Morgenstern’s development of **game theory** (see also Luce and Raiffa, 1957) included the surprising result that, for some games.
- **Operations research**, which emerged in World War II from efforts in Britain to optimize radar installations, & later found civilian applications in complex management decisions.
- The pioneering AI researcher Herbert Simon (1916–2001) won the Nobel Prize in economics in 1978 for his early work showing that models based on **satisficing**—making decisions that are “good enough”.

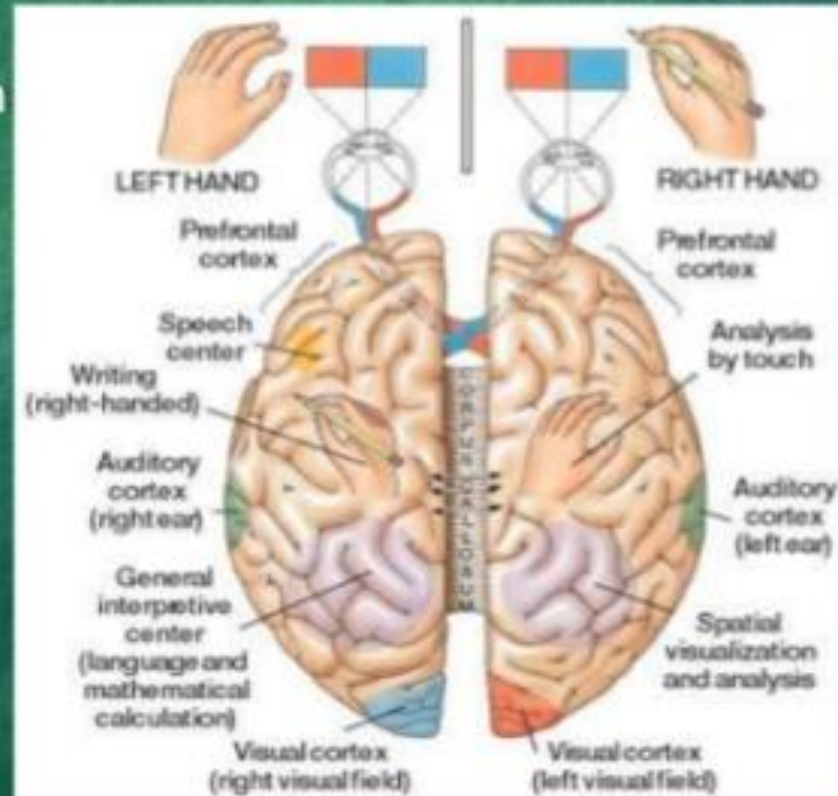
4) Neuroscience:-

- How do brains process information?
- **Neuroscience** is the study of the nervous system, particularly the brain.
- Paul Broca's (1824–1880) study of aphasia (speech deficit) in brain-damaged patients in 1861 demonstrated the existence of localized areas of the brain responsible for specific cognitive functions.
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- He showed that speech production was localized to the portion of the left hemisphere now called Broca's area.

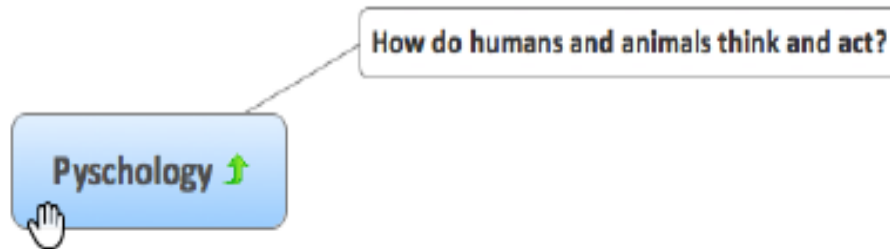
Neuroscience:-

Neuroscience

- Certain portions of the brain process information
- Other portions may store the information
- It stores information in axons and transmits through the synapsis
- Unused sections of the brain atrophy
- Experience and environment alter the brain



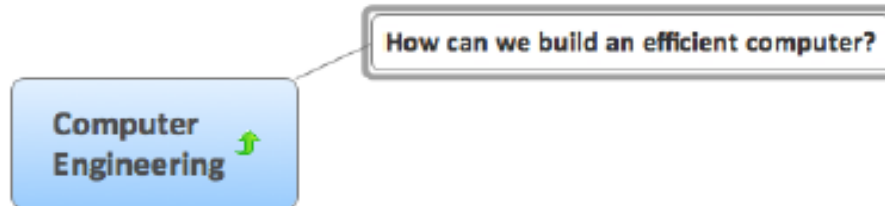
5) Psychology



- In 1940's concepts of mental “beliefs” and “goals” established – key items in AI today
 - ▶ Example is that pressure and temperature are used to discuss gases...
 - ▶ ...despite consisting of molecules that have neither
- Cognitive psychology views brain as information-processing device
- Cognitive science based on computational models of “thinking” – addressing psychology of memory, language, and logical thinking

6) Computer Engineering

AI Foundations...Computer Engineering



- For artificial intelligence to succeed needed two things – intelligence and an artifact - computer represents latter
- Computers were simultaneously developed by three countries embattled in World War II
 - ▶ Though early models specific in focus (deciphering German messages)...
 - ▶ ...later models have evolved rapidly – performance doubling every 18 months
- AI has benefitted from computer hardware and software advancement...
- ...in turn computers have benefitted from AI in terms of time sharing , interactive interpreters, symbolic and object programming

7) Control theory and Cybernetics

AI Foundations...Control Theory & Cybernetics



- **Ktesibios (c. 250 BC) built first self-controlling machine**
 - ▶ Water clock with regulator maintaining constant flow rate
 - ▶ Invention changed the definition of what an artifact could do
- **Norbert Wiener (1894-1964) viewed purposeful behavior arising from a regulatory mechanism trying to minimize “error”...**
- **...and his work spawned new mathematical and computational cognition models**
- **Modern control theory – especially stochastic optimal control – has goal of systems that maximize objective function over time...**
- **...matching view of AI – designing systems that behave optimally**

8) Linguistics- use to study languages.

AI Foundations...Linguistics



- Noam Chosky published *Syntactic Structures* in 1950's – based on syntactic models by Indian linguist Panini (c. 350 BC)
 - ▶ Addressed the creativity in language
 - ▶ For example, key question is how a child could understand and make up sentences that he or she have never heard before
 - ▶ Chosky's models were formal enough that they could be programmed
- Modern linguistics and AI were “born” at the same time and led to field of Natural Language Processing
- Understanding language is particularly difficult AI problem
 - ▶ Knowledge of sentence structure is not enough
 - ▶ Knowledge of subject matter and context also required
 - ▶ Much of early Knowledge Representation work was tied to language

THE HISTORY OF ARTIFICIAL INTELLIGENCE:-

1) The gestation of artificial intelligence (1943–1955)

- The first work that is now generally recognized as AI was done by Warren McCulloch and Walter Pitts (1943).
- They drew on three sources: knowledge of the basic physiology and function of neurons in the brain; a formal analysis of propositional logic due to Russell and Whitehead; and **Turing's theory** of computation.
- They proposed a model of artificial neurons in which each neuron is characterized as being “on” or “off,” with a switch to “on” occurring in response to stimulation by a sufficient number of neighboring neurons.
- Norbert Wiener, who was promoting analog cybernetic devices rather than digital computers.
- Now Carnegie Mellon University (CMU).
- Newell & Simon also invented a list-processing language, IPL to write LT. They had no compiler and translated it into machine code by hand. To avoid errors, they worked in parallel, calling out binary numbers to each other as they wrote each instruction to make sure they agreed.

- McCulloch & Pitts: Boolean circuit model of brain.
- Donald Hebb (1949) demonstrated a simple updating rule for modifying the connection strengths between neurons. His rule, now called **Hebbian learning**, remains an influential model to this day.
- Two undergraduate students at Harvard, Marvin Minsky and Dean Edmonds, built the first neural network computer in 1950. The SNARC, as it was called, used 3000 vacuum tubes and a surplus automatic pilot mechanism from a B-24 bomber to simulate a network of 40 neurons.
- Alan Turing's introduced Turing test.

2) The birth of artificial intelligence (1956)

John McCarthy coined the term *Artificial Intelligence*. Demonstration of the first running AI program at Carnegie Mellon University.

- This was the first official usage of McCarthy's term *artificial intelligence*. Perhaps "computational rationality" would have been more precise and less threatening, but "AI" has stuck. At the 50th anniversary of the Dartmouth conference, McCarthy stated that he resisted the terms "computer" or "computational" .

The birth of artificial intelligence (1956)

- John McCarthy in summer of 1956 organized two-month workshop at Dartmouth – that became official birthplace of AI – proposing:
 - ▶ *Two Month, Ten man study of artificial intelligence (a term he coined)*
 - ▶ *Based on conjecture that every aspect of learning or other feature of intelligence can in principle be so precisely described that a machine can simulate it*
 - ▶ *Attempt will be made to find how to make machines use language, form abstractions and concepts...*
 - ▶ *...solving problems now reserved for humans,...*
 - ▶ *...and improve themselves*
- Two researches – Allen Newell and Herbert Simon – “stole the show” already had a reasoning program
- Dartmouth workshop did not lead to any new breakthroughs, but it introduced all of the major players to each other

3) Early enthusiasm, great expectations (1952–1969):--

The early years of AI were full of successes-in a limited way. General Problem Solver(GPS) was a computer program created in 1957 by Herbert Simon and Allen Newell to build a universal problem solver machine. The order in which the program considered sub goals and possible actions was similar to that in which humans approached the same problems.

Thus, GPS was probably the first program to embody the "thinking humanly" approach. At IBM, Nathaniel Rochester & his colleagues produced first AI programs.

Herbert Gelernter (1959) constructed the Geometry Theorem Prover, which was able to prove theorems that many students of mathematics would find quite tricky.

Lisp was invented by John McCarthy in 1958 while he was at the Massachusetts Institute of Technology (MIT).

In 1963, McCarthy started the AI lab at Stanford.

Tom Evans's ANALOGY program (1968) solved geometric analogy problems that appear in IQ tests.

4) A dose of reality (1966–1973):-

From the beginning, AI researchers were not shy about making predictions of their coming successes. The following statement by Herbert Simon in 1957 is often quoted: “It is not my aim to surprise or shock you-but the simplest way I can summarize is to say that there are now in the world machines that think, that learn & that create. Moreover, their ability to do these things is going to increase rapidly until-in a visible future-the range of problems they can handle will be coextensive with the range to which the human mind has been applied.

5) Knowledge-based systems: The key to power? (1969–1979):--

Dendral was an influential pioneer project in artificial intelligence (AI) of the 1960s, and the computer software **expert system** that it produced. Its primary aim was to help organic chemists in identifying unknown organic molecules, by analyzing their mass spectra and using knowledge of chemistry. It was done at Stanford University by Edward Feigen Baum, Bruce Buchanan, Joshua Lederberg, and Carl Djerassi.

6) AI becomes an industry (1980–present):--

In 1981, the Japanese announced the "Fifth Generation" project, a 10-year plan to build intelligent computers running Prolog.

Overall, the AI industry boomed from a few million Dollars in 1980 to billions of dollars in 1988.

7) The return of neural networks (1986–present):--

Psychologists including David Rumelhart & Geoff Hinton continued the study of neural-net models of memory.

8) AI adopts the scientific method (1987–present)

- In recent years, approaches based on hidden Markov models (HMMs) have come to dominate the area.
- Speech technology and the related field of handwritten character recognition are already making the transition to widespread industrial and consumer applications. The Bayesian network formalism was invented to allow efficient representation of, and rigorous reasoning with, uncertain knowledge.

9) The emergence of intelligent agents (1995–present):--

One of the most important environments for intelligent agents is the Internet.

- Understanding agents working in real time environment using various inputs e.g recommender system, search engines website constructor etc.
- Reasoning and planning systems in addition to sensor inputs.
- Simon's words-"machines that think, that learn & that create." call it as human-level AI or HLAI.

10) The availability of very large data sets (2001–present):-

- Throughout 60 year history of computer science emphasis has been placed on the algorithm as main subject of study
- However, some recent AI suggests that for many problems it makes more sense to worry about the data...
- ...based on increasing availability of very large data sources – e.g., trillions of words on Web, billions of base pairs in human genome
- An example of this approach is interpreting the word “plant” as it relates to flora or a factory
 - ▶ Has been shown that with large unannotated data and minimal knowledge...
 - ▶ ...can bootstrap to learn new patterns that help label new examples
- This implies future AI systems that can learn on their own given enough data to work with

State of the art of AI:-

- **What can AI do today?**

- 1) **Robotics:-** A driverless robotic car, named STANLEY was the first to win the 2005 DARPA Grand Challenge. In 2006, CMU's BOSS won the Urban challenge, safely driving in traffic , obeying the traffic rules and avoiding pedestrians and other vehicles.
- 2) **Speech Recognition:-** Converts spoken words into text, helpful for note-taking or creating subtitles.
- 3) **Autonomous planning and scheduling:-** A hundred millions miles from earth, NASA's Remote Agent program became the first on-board autonomous planning program to control the scheduling of operations for a spacecraft.
- 4) **Game playing:-** Deep Blue defeated the reigning world chess champion Garry Kasparov in 1997, Proverb solves crossword puzzles better than most humans.

State of the art

5) Spam fighting:- deleting spam message and saving human times.

6) Logistics planning:- DART- Dynamic Analysis and Replanning Tool, do the automated planning and scheduling for transportations.

7) Machine translation:- Machine translation is done entirely by a computer program, Google Translate being unquestionably the most widely known and used example. Language professionals sometimes review machine translations in a process called post-editing (a cursory review to improve the final product).